Evaluating Mamba Models for Nonstationary Financial Forecasting: A Locked Multi-Panel Benchmark

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Interim protocol-aligned manuscript. Confirmatory test outcomes remain locked and are not reported in this version.

Abstract

Financial level series exhibit persistent levels, changing volatility, and evolving cross-series relationships, conditions under which model rankings can depend strongly on the evaluation protocol. We define a prospectively frozen benchmark of four daily financial panels: foreign exchange, U.S. Treasury yields, cross-asset indicators, and cryptoassets. Each panel yields a univariate-input task and a multivariate-input, single-target task. Nine learned models—five non-Mamba baselines, S-Mamba, and three decomposition or multi-scale Mamba variants—are compared with last-value, random-walk-with-drift, ARIMA, and, for multivariate tasks, VAR-on-differences references at horizons of 1, 5, 20, 60, 120, and 252 observations. Hyperparameters are selected using 2021–2022 validation data after training through 2020; the complete validation-selection audit has passed, while the 2023–2026 final period remains locked. Selection-safe diagnostics show that all four training targets are consistent with nonstationary levels under the prespecified ADF/KPSS diagnostic pair, while validation-period mean shifts range from 0.51 to 7.75 training standard deviations in magnitude and first-difference volatility ratios range from 0.82 to 4.52. Final neural estimates will average five seeds, and paired loss contrasts will use a moving-block bootstrap with multiplicity adjustment. Because the locked evaluation has not been opened, this interim manuscript reports the protocol and data characterization but makes no performance claim.

1 Introduction

Financial forecasting is frequently evaluated on price or rate levels that combine strong persistence with changes in location, scale, volatility, and dependence across markets. These properties complicate both modeling and evaluation. A flexible model can appear accurate by reproducing persistence, while a random chronological split can expose it to a future period unlike the one used for fitting. Accordingly, evidence for a forecasting architecture requires explicit reference forecasts,

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fixed temporal boundaries, selection separated from final evaluation, and uncertainty estimates that respect dependent forecast origins [1, 2, 3].

Modern long-horizon forecasters use linear decompositions, temporal patches, two-dimensional temporal representations, or inverted variable tokens [4, 5, 6, 7]. Mamba provides an input-selective state-space operator with sequence-length scaling that is linear in the recurrent scan [8]. Forecasting adaptations differ in whether the scan acts across variables or temporal tokens and whether decomposition or multiple temporal resolutions are introduced [9, 10, 11, 12, 13, 14, 15]. These architectural differences are plausible responses to changing temporal structure, but plausibility alone does not show that they improve out-of-period financial forecasts.

This study asks three bounded questions. First, do the tested Mamba variants improve level forecasts relative to persistence, drift, statistical models, and tuned non-Mamba neural baselines? Second, are any differences stable across foreign-exchange, yield-curve, cross-asset, and cryptoasset panels and across forecast horizons from one observation to approximately one trading year? Third, does adding contemporaneously available panel histories improve single-target forecasts relative to the paired univariate-input task? The study predicts statistical levels rather than returns and does not evaluate trading profitability.

To answer these questions without reusing the development evidence, we froze version 1 of the financial confirmatory protocol before opening its final period. Four panels generate eight paired tasks: S mode uses only the target history, whereas MS mode supplies the complete panel history but evaluates only the named target. Nine learned models share a 12-trial selection budget per task and are evaluated at six horizons using five final seeds. The final period, 2023-01-01 through 2026-06-30, is protected by an explicit software gate until all validation selections and their hashes pass audit. Legacy GLD and Exchange experiments informed development and are excluded from confirmatory claims; non-financial application benchmarks are outside the frozen scope.

The contributions of the study design are:

- a checksum-bound, fixed-calendar benchmark spanning four financial targets, paired S/MS tasks, six horizons, and five final seeds;
- an implementation-grounded comparison of inverted, decomposition, multi-scale, and patch-temporal Mamba configurations against neural and statistical references;
- a prespecified analysis of path and terminal errors, improvement relative to last-value, paired S/MS differences, volatility regimes, and dependence-aware bootstrap contrasts; and
- fail-closed result generation that withholds final tables until every required model, seed, task, horizon, and provenance field is present.

This version is an interim, protocol-aligned manuscript. It reports only training/validation data characterization; performance results and conclusions remain deliberately unwritten until the locked matrix is complete.

2 Related Work

2.1 Long-Horizon Forecasting

Neural forecasting architectures encode temporal structure in substantially different ways. Informer and Reformer reduce the computational cost of attention [16, 17]; Autoformer and FEDformer incorporate decomposition [18, 19]; and DLinear shows that a small decomposition-based linear model can remain competitive [4]. PatchTST uses temporal patches, TimesNet maps temporal

variation into two-dimensional representations, and iTransformer represents variables rather than time points as tokens [5, 6, 7]. N-BEATS and N-HiTS provide further examples of basis expansion and explicit multi-rate structure without attention [20, 21]. These alternatives motivate a diverse comparison rather than one restricted to Transformer variants.

2.2 Nonstationarity and Distribution Shift

Forecasting under distribution shift has motivated both normalization and architecture-level responses. RevIN normalizes each input instance and reverses the transformation at the output [22]. Non-stationary Transformers argue that normalization can remove predictive information and introduce a learned de-stationary mechanism [23]. Decomposition methods separate trend-like and residual structure so that one representation need not explain all temporal scales [18, 19, 4]. The variants evaluated here combine reversible window normalization with decomposition, multiple downsampling rates, or temporal patches, but their component labels denote architectural constructions rather than uniquely identified financial processes.

2.3 Mamba for Time Series

Mamba replaces self-attention with an input-selective state-space recurrence [8]. S-Mamba applies bidirectional Mamba blocks over variable tokens for forecasting [9]; Bi-Mamba+ studies bidirectionality and token relations [10]; MambaTS examines temporal and variable scan strategies [11]; and TimeMachine combines multiple Mamba pathways under channel-mixing and channel-independent settings [12]. TSMamba further explores transfer through a bidirectional time-series foundation model [24].

Several later studies overlap directly with the decomposition and multi-scale ideas considered in this repository. ms-Mamba uses Mamba blocks operating at multiple temporal sampling rates [13]; KARMA combines multilevel decomposition with multi-scale Mamba blocks for multivariate forecasting [14]; and DMamba assigns different forecasting modules to decomposed components [15]. The present study does not claim priority for decomposition or multi-scale Mamba as general principles. Its narrower purpose is to compare four implemented configurations under a prospectively frozen financial protocol.

2.4 Financial Forecast Evaluation

Financial time-series studies are sensitive to the definition of the target, temporal split, regime coverage, and reference forecast [3]. For persistent levels, a last-value forecast is an essential benchmark; drift and classical univariate or multivariate models provide additional checks that apparent neural gains are not consequences of an undemanding comparator. Forecasting competitions likewise show that rankings vary by series and horizon and that simple methods remain competitive [1, 2].

The present benchmark therefore reports each task and horizon rather than pooling raw errors across targets with different units. It uses a fixed future period, repeated neural training, and paired loss comparisons over common forecast origins. Overlapping origins are handled with moving-block resampling rather than treated as independent observations. These choices estimate performance within the frozen panels and dates; they do not identify a stable market mechanism or establish economic value.

3 Forecasting Models

3.1 Paired Forecasting Tasks

Let $\mathbf{X}_{t-L+1:t} \in \mathbb{R}^{L\times C}$ denote the observations available at forecast origin t, and let $Y_{t+1:t+H} \in \mathbb{R}^{H}$ denote the future path of one target series. For each financial panel, the S task sets $C = 1$ and supplies only the target history. The paired MS task supplies all $C > 1$ panel variables, with the target stored in the last channel. Although MS models emit one forecast per input channel internally, training loss and evaluation select only this final target channel. Thus, MS means multivariate input to single-target prediction rather than unrestricted multivariate forecasting. The horizon set is

$$H \in \{1, 5, 20, 60, 120, 252\}. \tag{1}$$

A separate model is fitted for every task and horizon; its lookback length L and other hyperparameters are selected using validation data only.

3.2 Selective State-Space Backbone

A discretized linear state-space model updates a latent state $\mathbf{h}_t$ and produces an output $\mathbf{y}_t$ as

$$\mathbf{h}_t = \bar{\mathbf{A}}\mathbf{h}_{t-1} + \bar{\mathbf{B}}\mathbf{x}_t, \qquad \mathbf{y}_t = \mathbf{C}\mathbf{h}_t. \tag{2}$$

Mamba makes the discretization and readout input-dependent, producing selective parameters $\bar{\mathbf{A}}_t$, $\bar{\mathbf{B}}_t$, and $\mathbf{C}_t$ [8]:

$$\mathbf{h}_t = \bar{\mathbf{A}}_t\mathbf{h}_{t-1} + \bar{\mathbf{B}}_t\mathbf{x}_t, \qquad \mathbf{y}_t = \mathbf{C}_t\mathbf{h}_t. \tag{3}$$

The repository encoder applies independent forward and reverse Mamba scans, sums their outputs, and uses residual connections, layer normalization, and a feed-forward sublayer. Figure 1 shows this conceptual block.

3.3 S-Mamba

S-Mamba uses reversible normalization and inverted variable tokens [9]. For sample b and variable c,

$$\mu_{b,c} = \frac{1}{L}\sum_{t=1}^{L} X_{b,t,c}, \qquad \sigma_{b,c} = \sqrt{\frac{1}{L}\sum_{t=1}^{L}(X_{b,t,c} - \mu_{b,c})^2 + \epsilon}, \tag{4}$$

$$Z_{b,t,c} = \frac{X_{b,t,c} - \mu_{b,c}}{\sigma_{b,c}}. \tag{5}$$

The entire normalized history of variable c is projected to one token,

$$\mathbf{e}_{b,c} = W_{\text{emb}}\mathbf{z}_{b,:,c} + \mathbf{b}_{\text{emb}}, \qquad \mathbf{E} \in \mathbb{R}^{B\times C\times d_{\text{model}}}. \tag{6}$$

After bidirectional Mamba encoding, a linear head produces H values per variable and the window transformation is reversed. In S tasks, the encoder receives one variable token and therefore cannot model cross-series relations. In MS tasks, it receives all panel variables and can represent associations among their histories, although the paired design does not establish that any such association is causal.

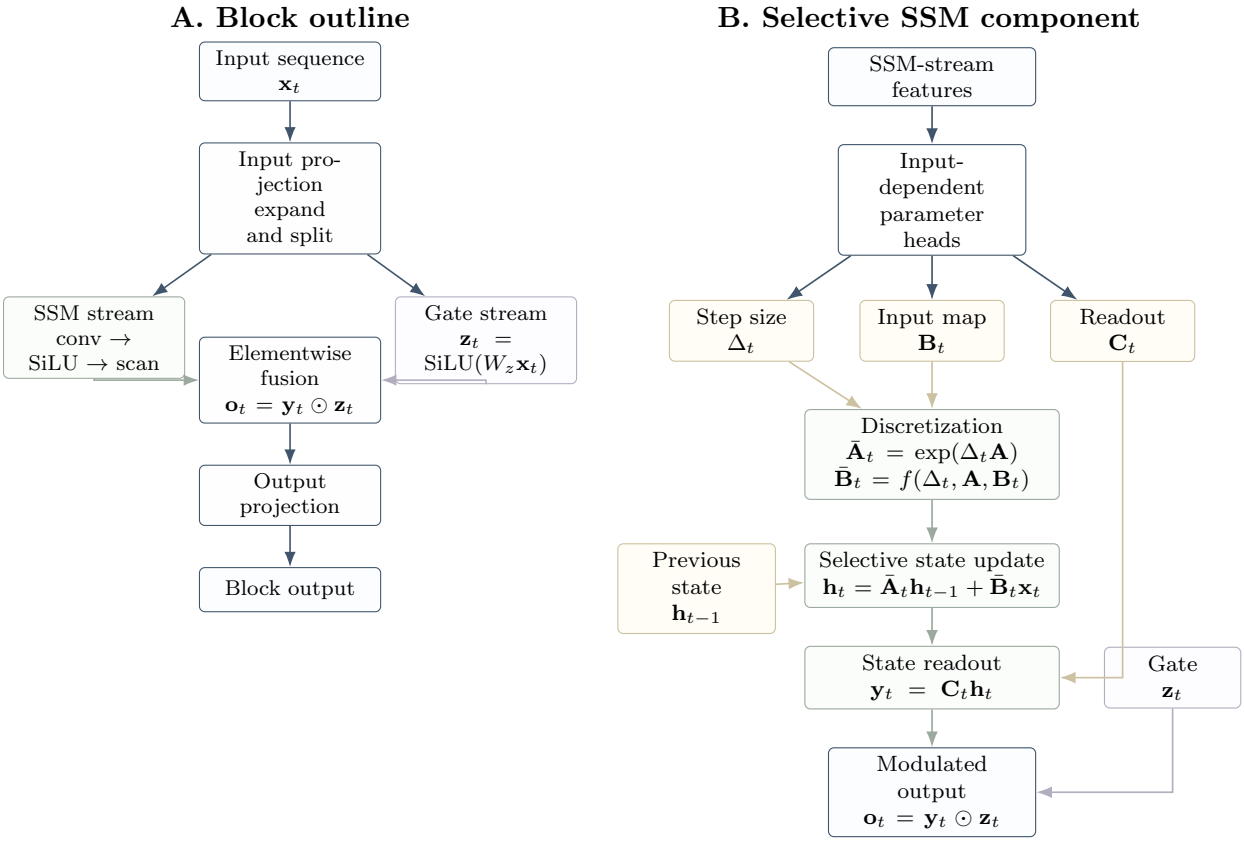


Figure 1: Conceptual bidirectional Mamba block used by the forecasting encoders. The diagram describes the implemented information flow; it does not imply a causal interpretation of the learned state.

3.4 Decomposition and Multi-Scale Variants

NS-Mamba-D uses standard reversible normalization and a three-way moving-average decomposition. With centered moving average $\mathrm{MA}_m(\cdot)$ and boundary replication,

$$\mathbf{Z}^{\mathrm{tr}} = \mathrm{MA}_m(\mathbf{Z}), \tag{7}$$

$$\mathbf{R}^0 = \mathbf{Z} - \mathbf{Z}^{\mathrm{tr}}, \tag{8}$$

$$\mathbf{Z}^{\mathrm{se}} = \mathrm{MA}_m(\mathbf{R}^0), \tag{9}$$

$$\mathbf{Z}^{\mathrm{re}} = \mathbf{R}^0 - \mathbf{Z}^{\mathrm{se}}. \tag{10}$$

Independent S-Mamba branches forecast the three components, and their normalized forecasts are summed before inverse normalization. The labels trend, seasonal, and residual name architectural components only; they are not identified economic processes.

MS-Mamba-MD first concatenates the three components along the variable dimension and forms average-pooled temporal scales $s \in \mathcal{S}$:

$$\mathbf{V}_s = \mathcal{P}_s\big(\mathrm{concat}_C[\mathbf{Z}^{\mathrm{tr}}, \mathbf{Z}^{\mathrm{se}}, \mathbf{Z}^{\mathrm{re}}]\big)\,. \tag{11}$$

One inverted-token Mamba branch processes the $3C$ component tokens at each scale. Its component forecasts are summed within variable, and the frozen MD configuration averages forecasts over scales.

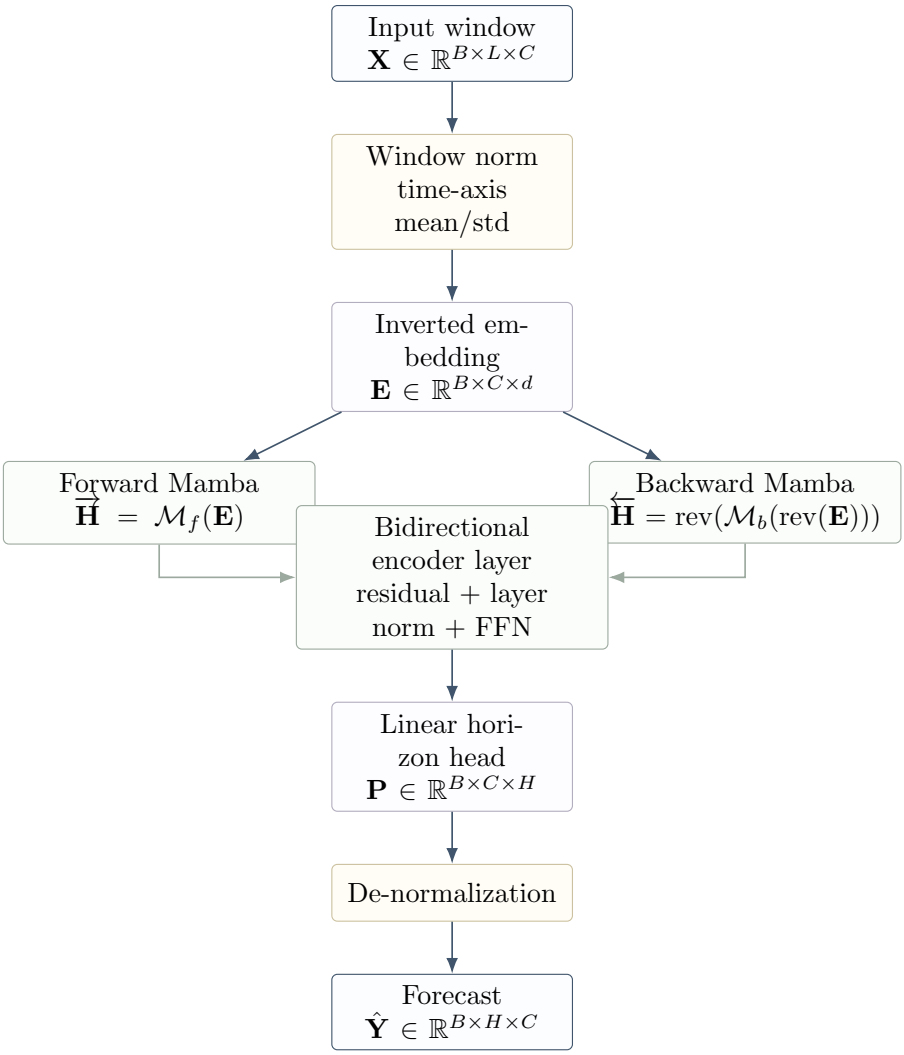


Figure 2: S-Mamba forecasting encoder with reversible normalization, variable-token embedding, bidirectional encoding, horizon projection, and inverse normalization.

The patch variant, PMS-Mamba-PD, retains the decomposition and scale construction. It replaces each inverted-token branch with temporal patch tokens. For component channel k, patch length p, and stride r, patch j is embedded as

$$\mathbf{e}_{j,k}^{(s)} = W_p \,\mathrm{vec}(\mathbf{V}_{s,j:j+p-1,k})\,. \tag{12}$$

Mamba scans the patch sequence, a flattened head maps it to H values, and component and scale forecasts are combined by sums and means, respectively. Table 1 summarizes the fixed distinctions among the four Mamba labels.

3.5 Learned and Statistical Comparators

The learned comparison set contains DLinear, NLinear, PatchTST, TimesNet, and iTransformer [4, 5, 6, 7]. These models cover decomposition-based and last-value-centered linear mappings, temporal patch attention, two-dimensional temporal modeling, and inverted variable-token attention. All learned models use the same task-specific optimization budget and final seed set.

Table 1: Mamba configurations in the frozen benchmark.

Variant	Tokenization and structure	Fusion
S-Mamba	Inverted variable tokens	Single path
NS-Mamba-D	Three independent decomposition branches	Component sum
MS-Mamba-MD	Component tokens at multiple temporal scales	Mean over scales
PMS-Mamba-PD	Temporal patches at multiple decomposed scales	Mean over scales

Four non-neural references address the strong persistence of financial levels. Last-value repeats Y_t for every lead. Random-walk-with-drift extrapolates the average change from the first available observation through origin t. ARIMA order (p, d, q) is chosen on training data by minimum AIC over $d \in \{0, 1\}$ and $p, q \in \{0, 1, 2, 3\}$, excluding ARIMA$(0, 0, 0)$. For MS tasks only, VAR-difference selects a lag from 1 through 10 by training BIC, forecasts first differences of the complete panel, and accumulates the target differences back to levels. Orders and lags are selected using training data only. Coefficients are then estimated once from all pre-test training and validation observations; states are updated sequentially with newly observed test values without refitting coefficients at every origin.

3.6 Training Objective

All learned models minimize mean squared error on the scaled target channel:

$$\mathcal{L} = \frac{1}{BH} \sum_{b=1}^{B} \sum_{h=1}^{H} \left(\hat{Y}_{b,h} - Y_{b,h} \right)^2 . \tag{13}$$

Each column is standardized using training data only. Predictions and targets are transformed back to the target's original scale before evaluation.

4 Experimental Setup

4.1 Financial Panels and Data Provenance

The frozen data cutoff is 2026-06-30. Table 2 describes four daily panels after chronological sorting and complete-case intersection. The foreign-exchange panel uses six FRED exchange-rate series, with reciprocal transformations where needed so that every column is quoted as U.S. dollars per unit of foreign currency [25]. The yield panel contains 2-, 5-, 10-, and 30-year U.S. Treasury constant-maturity rates from FRED, with DGS10 as target [25]. The cross-asset panel aligns FRED S&P 500, WTI crude-oil, and VIX levels with the unadjusted GLD close obtained from Yahoo Finance [25, 26]. The cryptoasset panel uses daily UTC spot-closing prices from the official Binance public archive [27]. For the latter two panels, redistribution rights must be confirmed before a public data release; the local source archives, URLs, retrieval dates, byte counts, and SHA-256 hashes are retained.

All processed panels have unique increasing dates and no missing cells. Complete-case alignment avoids imputed targets but produces an irregular observation calendar governed by common market availability; horizons are therefore measured in panel observations, not elapsed calendar days. The fixed splits are training through 2020-12-31, validation from 2021-01-01 through 2022-12-31, and final evaluation from 2023-01-01 through 2026-06-30. Validation and test loaders can use preceding

Table 2: Frozen financial panels. Counts refer to target observations within each calendar split. Each panel defines a target-only S task and a complete-panel MS task.

Panel	Date range	Inputs in MS mode (target last)	Target	Total	Train	Validation	Test
Foreign exchange	1999-01-04–2026-06-30	EURUSD, CADUSD, JPYUSD, CHFUSD, AUDUSD, GBPUSD	GBPUSD	6,894	5,521	499	874
Treasury yields	2006-02-09–2026-06-30	DGS2, DGS5, DGS30, DGS10	DGS10	5,101	3,728	500	873
Cross asset	2016-07-25–2026-06-30	S&P 500, WTI, VIX, GLD	GLD	2,483	1,112	502	869
Cryptoasset	2018-05-04–2026-06-30	ETHUSDT, BNBUSDT, XRPUSDT, BTCUSDT	BTCUSDT	2,980	973	730	1,277

Table 3: Selection-safe diagnostics computed without reading final-period targets. Mean shift is measured in training standard deviations; standard-deviation and difference-volatility columns are validation-to-training ratios. Correlation shift is the largest absolute change in a feature–target first-difference correlation.

Panel	Level ADF p	Level KPSS p	Mean shift	SD ratio	Difference-volatility ratio	Max. correlation shift
Foreign exchange	0.682	≤ 0.010	-1.294	0.430	0.823	0.268
Treasury yields	0.363	≤ 0.010	-0.512	0.858	1.129	0.026
Cross asset	0.719	≤ 0.010	1.708	0.344	1.251	0.323
Cryptoasset	1.000	≤ 0.010	7.750	3.669	4.521	0.106

observations as input context, but every forecast target remains inside its named split. Each scaler is fitted only on the training rows.

The legacy GLD and Exchange benchmarks were used during model development and are excluded from the present evidence boundary. Non-financial application benchmarks are also outside the protocol because the research question is specifically nonstationary financial forecasting rather than broad application-domain ranking. These exclusions were frozen with the new protocol and are not based on outcomes from its protected final period.

4.2 Selection-Safe Nonstationarity Characterization

Before test access, diagnostics read only training and validation targets. For training levels, the augmented Dickey–Fuller (ADF) test includes a constant and trend, and the KPSS test uses trend stationarity as its null [28, 29]. The same tests are applied to first differences with a constant but no trend. The diagnostic pair is interpreted descriptively: failure to reject an ADF unit-root null together with rejection of the KPSS trend-stationarity null supports treating the observed level as nonstationary, but neither test identifies a unique data-generating mechanism.

Table 3 reports the prespecified level tests and validation shifts. All four training targets show ADF $p > 0.36$ and KPSS $p \leq 0.01$ for levels. The validation period also changes materially in at least one location, scale, difference-volatility, or cross-series-correlation summary for every panel. Cryptoassets show the largest location and difference-volatility shifts, while cross-asset and foreign-exchange panels show the largest maximum absolute changes in feature–target first-difference correlations.

Figure 3 visualizes the same evidence boundary. Targets are standardized by training means and standard deviations; trailing 60-observation first-difference volatility is divided by its training median. The gray interval is validation, and no 2023–2026 target is shown.

4.3 Selection and Locked Evaluation

The learned set contains DLinear, NLinear, PatchTST, TimesNet, iTransformer, S-Mamba, NS-Mamba-D, MS-Mamba-MD, and PMS-Mamba-PD. Every model–task study receives 12 effective Optuna trials, where completed and pruned trials count toward the fixed budget. A seeded TPE

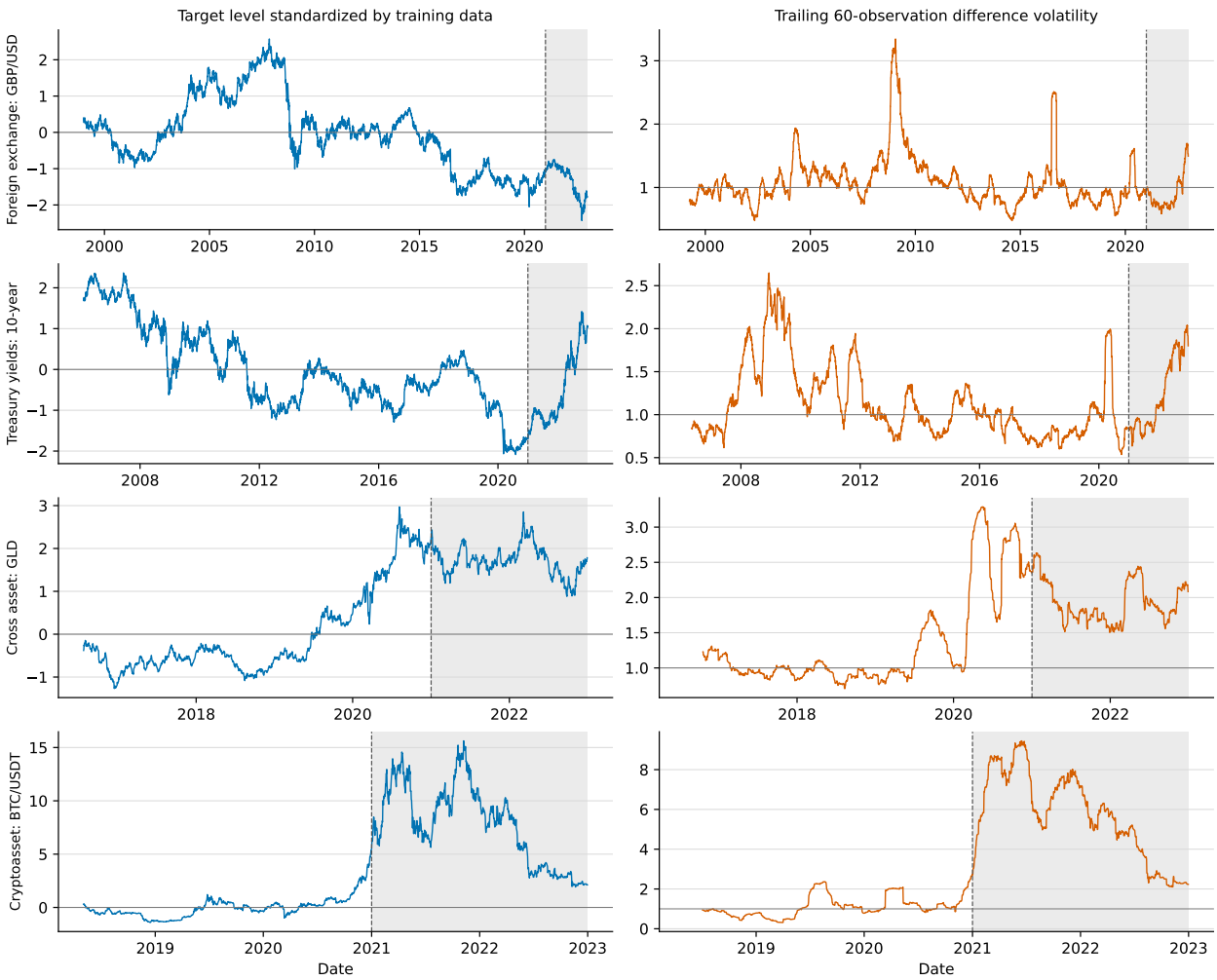


Figure 3: Selection-safe financial data characterization. Dashed vertical lines mark the start of validation, and gray regions denote validation observations. The left column shows target levels standardized using training data only. The right column shows lagged trailing 60-observation first-difference volatility relative to its training median. Test targets are neither read nor plotted.

sampler evaluates calibration horizons 5, 60, and 252; median pruning begins after four startup trials. The three best completed calibration trials are then replayed across all six validation horizons. The configuration with the smallest balanced validation RMSSE is frozen for that model and task. Training uses Adam, at most 10 epochs, early-stopping patience 3, and mean squared error. Model-specific search spaces are listed in Appendix A.1.

No final-period evaluation is permitted during selection. The selection audit verifies task and model coverage, protocol and data hashes, horizon lists, validation flags, and that the final-evaluation flag remains unopened for every selected record. Only after this audit may the explicit `--unlock-test` gate be invoked. The selected neural configurations are then retrained and evaluated with seeds 2023–2027. Neural weights are fitted on training observations, with validation observations used for early stopping rather than gradient updates. The complete neural matrix contains $8 \times 9 \times 5 = 360$ task–model–seed cells, each comprising six separately fitted horizon models. Statistical references refuse to access the final period until all 360 neural cells are complete.

4.4 Metrics and Prespecified Analysis

For task d, horizon H, forecast origin o, and lead h, let $e_{d,o,h} = Y_{d,o,h} - \hat{Y}_{d,o,h}$. Path MSE averages $e^2_{d,o,h}$ over every available origin and all leads $1, \ldots, H$; terminal MSE uses only lead H. MAE and RMSE are defined analogously. RMSSE divides MSE by the mean squared one-step difference of the inverse-transformed training target and then takes the square root [2]:

$$\mathrm{RMSSE}_{d,H} = \sqrt{\frac{\mathrm{MSE}_{d,H}}{(T_d - 1)^{-1} \sum_{t=2}^{T_d} (Y_{d,t} - Y_{d,t-1})^2}}. \tag{14}$$

Metrics remain task-specific; raw errors are never averaged across targets with different units. Balanced RMSSE is the unweighted mean over the six horizons, equivalently the mean of the short $(1, 5)$, middle $(20, 60)$, and long $(120, 252)$ group means.

For each fitted model, skill relative to last-value is

$$\mathrm{Skill}_{d,H} = 1 - \frac{\mathrm{MSE}^{\mathrm{model}}_{d,H}}{\mathrm{MSE}^{\mathrm{LV}}_{d,H}}. \tag{15}$$

Positive values indicate lower path MSE than persistence. Neural metrics are summarized as mean and standard deviation over five seeds. S and MS estimates are paired by panel, model, and horizon.

For confirmatory contrasts, origin-level squared losses are first averaged across the five neural seeds. Each proposed variant (NS-Mamba-D, MS-Mamba-MD, and PMS-Mamba-PD) is paired with S-Mamba and with the non-Mamba model selected on validation for the same task. A circular moving-block bootstrap resamples the paired origin-level loss difference 10,000 times. The block length is $\min\{n, \max(H, \lceil n^{1/3} \rceil)\}$, and percentile 95% intervals are reported. Within each task, proposal, and comparator, two-sided bootstrap p-values across the six horizons receive Holm adjustment.

The prespecified secondary regime analysis assigns each origin using the standard deviation of target first differences over the preceding 60 observations, lagged by one observation. Training-period 0.25 and 0.75 quantiles define low-, middle-, and high-volatility regimes. Return forecasting is deferred until the primary level benchmark is complete and cannot replace an unfavorable level result.

4.5 Reproducibility and Evidence Boundary

Protocol version `financial_confirmatory_v1` was frozen on 2026-07-25. Its SHA-256 is

`1b1a8d94847ab2296dec0db9a5cbe596836aae15fd8e7d88c1e4082e178d5963`

Processed-data hashes, source hashes, split counts, software and GPU metadata, deterministic-algorithm flags, run times, and peak GPU memory are checked before manuscript generation. Interrupted Optuna studies resume from persistent storage, with stale running trials marked failed rather than silently counted as completed. Final tables cannot be generated from a partial task subset or from validation scores.

At the time of this manuscript version, selection remains in progress and the final outcome files do not exist. Accordingly, no validation ranking is presented as a result, no test target has been opened for manuscript analysis, and all performance statements remain pending.

5 Interim Results

5.1 Selection-Safe Evidence

The evidence presently available supports the intended nonstationary setting but not a forecasting-performance conclusion. Across all four training targets, the level ADF test fails to reject a unit root while the level KPSS test rejects trend stationarity at its reported lower p-value bound (Table 3). This concordance is not treated as proof that every series follows an integrated process; it establishes that a stationary-level assumption is not an adequate common description of the training panels.

The held-out validation interval differs visibly from training (Figure 3). Foreign exchange and Treasury yields shift downward by 1.294 and 0.512 training standard deviations, respectively, whereas GLD shifts upward by 1.708 standard deviations. BTCUSDT exhibits the largest shift: its validation mean is 7.750 training standard deviations above the training mean, its level standard deviation is 3.669 times the training value, and its first-difference standard deviation is 4.521 times the training value. These quantities justify evaluating distribution-shift-sensitive designs, but they do not predict which model will perform best in the untouched final period.

The MS tasks also face changing cross-series structure. The maximum absolute change from training to validation in a feature–target first-difference correlation is 0.268 for foreign exchange, 0.026 for yields, 0.323 for cross assets, and 0.106 for cryptoassets. Consequently, an MS improvement cannot be assumed merely because additional histories are supplied; the paired S/MS comparison is required.

5.2 Locked Outcome Status

Table 4 records the evidence boundary for this manuscript version. Source retrieval, preprocessing, checksums, fixed splits, selection-safe diagnostics, and validation-only hyperparameter selection are complete. The frozen selection audit has passed; the final neural matrix, statistical-reference forecasts, bootstrap contrasts, and full-period diagnostics have not been generated because the protected final period has not been opened. The result generator is designed to fail unless all required artifacts match the frozen protocol and data hashes.

Table 4: Evidence status for the interim manuscript. "Locked" means that the workflow requires an explicit gate and preceding completeness audit before reading final-period targets.

Evidence component	Status	Admissible use in this version
Source and processed-data manifests	Complete	Dataset provenance and counts
Training/validation diagnostics	Complete	Selection-safe characterization
Validation-only model selection	Complete (audit passed)	Configuration freeze only; no ranking claim
2023–2026 neural evaluation	Locked	None
Statistical-reference evaluation	Blocked by neural matrix	None
Bootstrap and regime outcome analysis	Not generated	None

5.3 Prespecified Final Reporting

After the protected evaluation is unlocked and validated, the final paper will report every task separately. The primary table will contain balanced path RMSSE for last-value, the validation-selected

non-Mamba comparator, S-Mamba, and the three proposed Mamba variants; neural entries will be mean ± standard deviation over five seeds. Horizon-specific path and terminal metrics, random-walk-with-drift, ARIMA, and VAR-difference estimates will be retained in generated supplementary tables rather than selectively omitted.

Claims of improvement will require more than the smallest point estimate. They will be based on paired origin-level loss differences, 95% moving-block bootstrap intervals, and Holm-adjusted p-values for the prespecified contrasts. The S/MS analysis will report signed differences for the same panel, model, and horizon. Regime results will remain secondary, and return forecasts will be labeled robustness analyses performed only after completion of the level benchmark.

6 Discussion and Limitations

The selection-safe evidence confirms that the benchmark poses the intended problem: persistent financial levels are accompanied by substantial temporal changes in location, scale, volatility, and, for some panels, feature–target correlations. The size and direction of these changes vary sharply across panels. This heterogeneity is precisely why the study is organized around task-level estimates and paired comparisons rather than a single rank averaged across unrelated units.

Several outcome interpretations are fixed in advance. First, a neural model is not considered useful merely because it outperforms another neural architecture; performance relative to last-value and other statistical references is central. Second, a favorable result on one panel or horizon does not establish general Mamba superiority. Third, an MS advantage indicates predictive value from the supplied panel histories under this protocol, not a causal cross-market mechanism. Fourth, level-forecast accuracy is not evidence of profitable trading, market inefficiency, directional accuracy, calibrated risk, or robustness to transaction costs.

The benchmark improves on the legacy development experiments by using a newly frozen final period, fixed task definitions, equal trial counts, five neural seeds, explicit statistical references, and dependence-aware paired resampling. The protocol and processed-data hashes are checked at every transition, and final manuscript generation is blocked until the complete matrix is available. These controls reduce opportunities for selective reporting, but they do not eliminate all researcher degrees of freedom. The model families and their search spaces reflect prior development, and the proposed Mamba variants are not accompanied by a full factorial ablation that isolates tokenization, decomposition, scale construction, and parameter count.

The data design also has limitations. Four panels cannot represent all financial markets, frequencies, or institutional regimes. Complete-case intersection removes imputation assumptions but changes each panel's observation calendar and may privilege dates on which every constituent market reports. The study uses one fixed final interval rather than multiple historical backtests, and the shortest cryptoasset training segment contains 973 observations. The final period may therefore provide a demanding distribution shift without supporting broad temporal generalization.

Statistical uncertainty remains conditional on the implemented resampling design. Forecast origins overlap, losses at different horizons are related, and five neural seeds quantify only part of optimization variability. The moving-block bootstrap preserves local origin dependence approximately; its block-length rule is prespecified rather than estimated for each series. Holm adjustment is applied within each task–proposal–comparator horizon family, not across every exploratory quantity in the paper. Regime estimates are secondary and may have unequal origin counts.

The statistical references are intentionally practical rather than exhaustive. ARIMA and VAR orders are selected on the training period; coefficients are estimated once from the full pre-test history, remain fixed during final evaluation, and observed values update the forecasting state

sequentially. In contrast, neural weights are estimated from training observations while validation controls early stopping. This asymmetric use of validation values gives the statistical references a conservative information advantage and must be retained when interpreting neural-to-statistical differences. More elaborate financial-econometric specifications, probabilistic forecasts, and models of conditional variance are outside the primary comparison. Specialist financial-econometrics review is therefore warranted before submission, particularly for any interpretation beyond point-forecast accuracy.

Finally, this interim version has no confirmatory performance result. Validation scores are model-selection evidence and will not be substituted for test outcomes. The discussion must be revised after the final matrix is complete, including explicit reporting of negative or null results. Dataset redistribution terms, a citable code archive, author contributions, funding, competing interests, and venue-specific AI-use disclosure also require accountable author confirmation before submission.

7 Conclusion

This work establishes a locked benchmark for evaluating Mamba-based forecasts of nonstationary financial levels. Four financial panels yield paired univariate-input and multivariate-input single-target tasks, evaluated at six horizons against five learned baselines and four Mamba configurations. The final design adds persistence, drift, ARIMA, and VAR-on-differences references, five neural seeds, paired S/MS analysis, moving-block bootstrap intervals, multiplicity adjustment, and prespecified volatility regimes.

Training/validation diagnostics show substantial nonstationarity and distribution shift, but they do not establish forecast superiority. The 2023–2026 final period remains locked after validation-only selection was completed and audited. Accordingly, the present conclusion is procedural rather than empirical: the study is positioned to test whether the proposed variants improve on strong financial references without reusing development outcomes, but that question remains unanswered in this manuscript version. Any final conclusion will be generated only from the complete validated evidence matrix and will preserve unfavorable, null, and panel-specific findings.

A Frozen Protocol and Reproducibility Details

A.1 Frozen protocol

Table 5: Controlling protocol fields.

Field	Frozen value
Version	`financial_confirmatory_v1`
Frozen on	2026-07-25
Data cutoff	2026-06-30
Train / validation / test	through 2020 / 2021–2022 / 2023–2026-06
Horizons	1, 5, 20, 60, 120, 252 observations
Calibration horizons	5, 60, 252 observations
Trials / full replays	12 effective trials / top 3 completed trials
Training cap / patience	10 epochs / 3 epochs
HPO sampler seed	20260725
Final neural seeds	2023, 2024, 2025, 2026, 2027
Bootstrap	10,000 circular moving-block replicates; seed 20260725

The eight task identifiers are listed by panel in Table 6.

Table 6: Paired task identifiers.

Panel	S task	MS task
Foreign exchange	`financial_fx_s`	`financial_fx_ms`
Treasury yields	`financial_yields_s`	`financial_yields_ms`
Cross asset	`financial_cross_asset_s`	`financial_cross_asset_ms`
Cryptoasset	`financial_crypto_s`	`financial_crypto_ms`

Every manifest entry stores its processed-data hash, the protocol hash, fixed split dates, target, input/output channel counts, horizon suite, and term groups.

A.2 Neural search spaces

The following values are categorical choices, not continuous ranges. All models use two encoder layers, dropout 0.1, and label length $L/2$ unless the implementation ignores a field. The common Mamba space is $L \in \{16, 32, 64, 96, 128, 192, 256, 320, 384\}$, $d_{\text{model}} \in \{64, 128\}$, $d_{\text{ff}} \in \{128, 256\}$, state dimension $\in \{16, 32\}$, moving-average window $\in \{13, 25, 49\}$, learning rate $\in \{5 \times 10^{-5}, 10^{-4}, 2 \times 10^{-4}, 3 \times 10^{-4}\}$, and batch size 16. S-Mamba adds no variant-specific choice. NS-Mamba-D fixes standard rather than adaptive normalization, activates decomposition, and disables shift gating. MS-Mamba-MD chooses scales from $\{1, 2, 4\}$, $\{1, 2, 4, 8\}$, $\{1, 3, 6\}$, or $\{1, 4, 8\}$; it fixes decomposition on, mean scale fusion, and no linear skip. PMS-Mamba-PD chooses $\{1, 2, 4\}$ or $\{1, 2, 4, 8\}$ and a patch-length/stride pair from $(8, 4)$, $(8, 8)$, $(16, 4)$, or $(16, 8)$; infeasible scaled patch layouts are rejected before training.

Optuna uses TPE with the frozen sampler seed and a median pruner with four startup trials, one warm-up step, and one-step intervals. Interrupted `RUNNING` trials are marked `FAIL` on recovery. The

Table 7: Non-Mamba categorical search spaces. "LR" denotes learning rate and "BS" batch size.

Model	Lookback L	Width choices	LR	BS	Specific choices
DLinear	96, 192, 256	–	0.003, 0.005	32, 64	Moving average 13, 25; shared channels
NLinear	64, 192, 320, 384	–	0.005, 0.01	32, 64	Shared channels
PatchTST	128, 256, 320	d_m: 128, 256; d_{ff}: 512, 2048	0.0002, 0.0003, 0.0005	16, 32	Patch/stride: 16/8, 24/12, 32/16
TimesNet	128, 256, 320	d_m: 32, 64; d_{ff}: 128, 256	0.0001, 0.0002, 0.0003	16, 32	Top-$k = 5$; kernels 6, 8
iTransformer	128, 192, 256	d_m: 128, 256; d_{ff}: 256, 512	0.0001, 0.0002, 0.0003	16	Reversible normalization on

fixed budget counts `COMPLETE` and `PRUNED` trials; failed trials do not consume it. Only completed trials can enter the top-three full validation replay.

A.3 Reference-model specification

Last-value and random-walk-with-drift have no fitted hyperparameters. ARIMA searches $d \in \{0, 1\}$ and $p, q \in \{0, 1, 2, 3\}$ on the training target and selects minimum AIC; ARIMA$(0, 0, 0)$ is excluded. A time trend is used for $d = 1$, and a constant plus trend is used for $d = 0$. VAR-difference is available only for MS tasks and selects lags 1–10 by minimum training BIC with constant and trend. After order or lag selection, coefficients are estimated once using all observations before the test boundary. At each final origin, ARIMA state and VAR lagged differences use observations available up to that origin, while fitted coefficients remain fixed.

A.4 Artifact gates and regeneration

The complete workflow is executed in the Anaconda environment with:

```
conda run --no-capture-output -n mamba python \
  scripts/financial/run_financial_benchmark.py --stage hpo
conda run --no-capture-output -n mamba python \
  scripts/financial/run_financial_benchmark.py --stage selection
conda run --no-capture-output -n mamba python \
  scripts/financial/run_financial_benchmark.py --stage final --unlock-test
conda run --no-capture-output -n mamba python \
  scripts/financial/run_statistical_baselines.py --unlock-test
conda run --no-capture-output -n mamba python \
  scripts/financial/analyze_financial_results.py
make -C manuscript financial-assets
```

The selection command refuses missing models, wrong hashes or horizons, non-validation selection, and prior test access. The final runner records task, model, seed, horizon, origin dates, predictions, targets, run time, peak GPU memory, software versions, GPU identity, and deterministic-algorithm settings. Completed cells are written atomically and skipped on restart.

The statistical runner is blocked until every neural cell is complete. The analysis script then verifies forecast shapes, origin dates, saved metrics, metadata, and expected row counts before writing aggregate metrics, S/MS pairs, regime metrics, and bootstrap contrasts. The manuscript generator additionally requires full-period diagnostics marked `test_targets_read=true` and exactly 10,000 bootstrap replicates. No partial task selection can produce the final result section.

The current interim paper instead uses only the selection-safe diagnostics manifest. That manifest records a training/validation scope and confirms that test targets were not read. The final generated tables and copied machine-readable supplements will be added only after the protected workflow completes.

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